Cybersecurity Home Lab

Setup, Network Configuration & First Nmap Port Scan

April 2026 | VirtualBox | Kali Linux | Metasploitable 2 | Nmap

1. Project Objective

The goal of this project was to build a fully isolated lab environment for practicing ethical hacking techniques. Using free virtualization software, I configured two virtual machines — an attacker machine (Kali Linux) and an intentionally vulnerable target (Metasploitable 2) — verified network connectivity between them, and performed an initial port and service scan using Nmap.

2. Tools and Technologies

Tool	Version/Type	Role
VirtualBox	Oracle (free)	Hypervisor for VMs
Kali Linux	Rolling release	Attacker machine
Metasploitable 2	Rapid7 (SourceForge)	Intentionally vulnerable target
Nmap	7.95	Network and port scanner

3. Virtual Network Configuration

To ensure all attack traffic remained completely isolated from the home network, I configured both VMs on a private internal network named **labnet**. Kali has two network adapters: one NAT adapter (for internet access) and one internal network adapter. Metasploitable only has the internal network adapter, preventing it from reaching the internet entirely.

Virtual Machine	Adapter 1	Adapter 2	Assigned IP
Kali Linux	NAT (internet)	Internal Network (labnet)	192.168.56.102
Metasploitable 2	Internal Network (labnet)	—	192.168.56.101

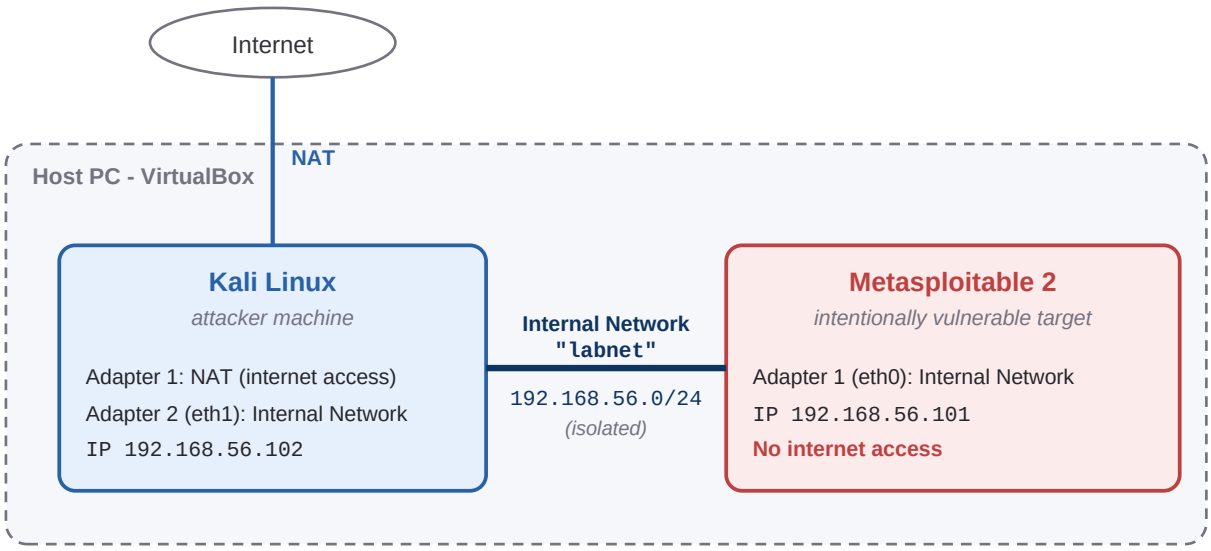


Fig. 1 — Lab network topology (schematic): only Kali can reach the internet; the target is reachable only through the isolated internal network

4. Metasploitable 2 Import

Metasploitable 2 is distributed as a **.vmdk** virtual disk file. I created a new VM in VirtualBox specifying Linux/Ubuntu 32-bit as the OS type, allocated 512 MB of RAM, and attached the .vmdk file as an existing virtual hard disk — no OS installation required.

5. Connectivity Verification

After booting both VMs, I manually assigned static IP addresses to the internal network adapters, since VirtualBox's internal network does not provide DHCP. Connectivity was then verified using ping.

Commands run on Metasploitable:

```
sudo ifconfig eth0 192.168.56.101 netmask 255.255.255.0
```

Commands run on Kali:

```
sudo ifconfig eth1 192.168.56.102 netmask 255.255.255.0
ping 192.168.56.101
```

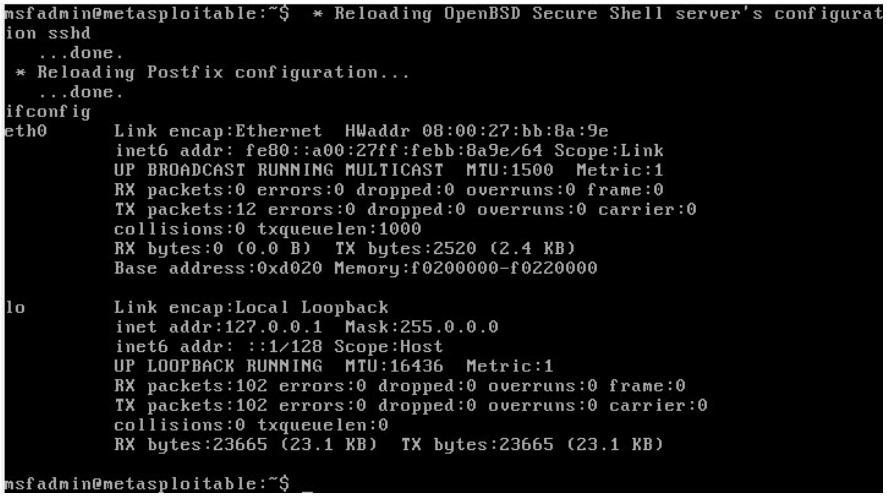


Fig. 2 — ifconfig output on Metasploitable after boot: eth0 is up but has no IPv4 address yet (expected, since the internal network has no DHCP)

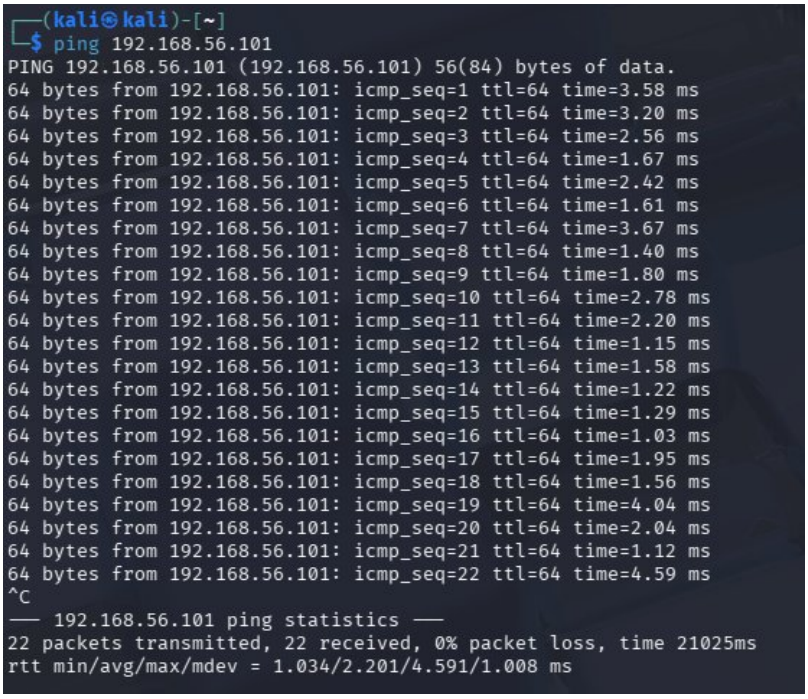


Fig. 3 — Ping from Kali to Metasploitable: 22 packets transmitted, 22 received, 0% packet loss

6. Nmap Port and Service Scan

With connectivity confirmed, I ran a default Nmap scan against Metasploitable to enumerate open ports and exposed services. A default scan checks the 1,000 most common TCP ports and does not include version detection or scripts. The command used was:

```
nmap 192.168.56.101
```

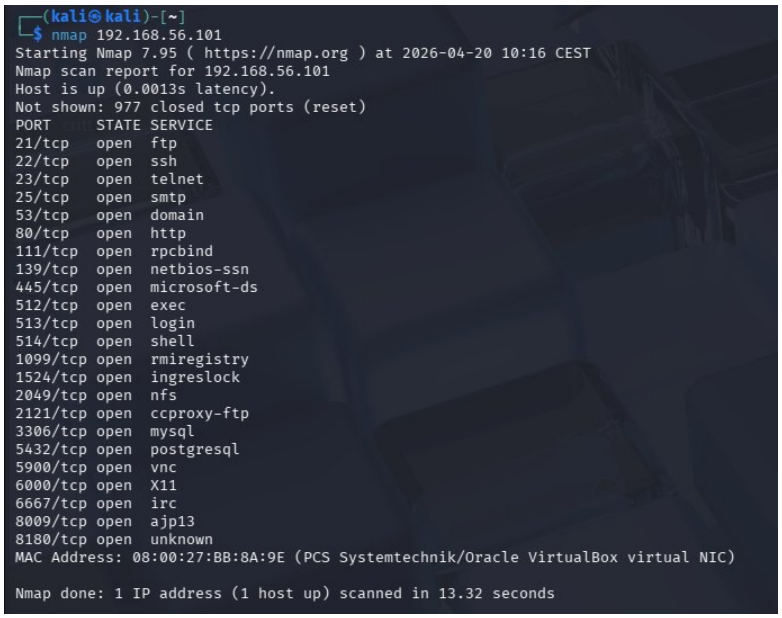


Fig. 4 — Nmap scan results: 23 open ports (977 closed) identified on Metasploitable, scan completed in 13.32 seconds

Selected open ports and potential risk:

Port	Service	Potential risk	Why it matters / what to check next
21/tcp	FTP	High	Cleartext file transfer; check for anonymous access and service version
22/tcp	SSH	Medium	Remote access; test credential strength in the lab
23/tcp	Telnet	High	Cleartext remote shell; credentials travel unencrypted
80/tcp	HTTP	Medium	Web server exposed; enumerate the hosted applications
1524/tcp	ingreslock	High	Port historically abused for backdoor shells; check what is listening
3306/tcp	MySQL	High	Database service reachable over the network
5900/tcp	VNC	High	Remote desktop service reachable
6667/tcp	IRC	Medium	Unnecessary service on a server

Risk levels are an initial assessment based on the type of service exposed, not confirmed vulnerabilities. A default scan does not identify service versions, so the next step is a scan with version detection and default scripts (`-sV -sC`) to verify each finding.

7. Conclusions & Next Steps

The home lab was successfully configured. Both virtual machines communicate reliably within an isolated environment, ready for safe ethical hacking practice. The default Nmap scan revealed 23 open ports on Metasploitable, showing a large attack surface to investigate.

Planned next steps:

- Re-run Nmap with version detection and default scripts (-sV -sC) to verify the services
- Test FTP anonymous login (port 21)
- SSH brute force attack with Hydra (port 22)
- Enumerate the web applications on port 80 (e.g., DVWA)
- Use Metasploit Framework for automated exploitation
- Document each attack as a writeup on GitHub

This project was carried out for educational purposes in a fully isolated environment. All activities were performed on virtual machines owned by the author.